

EXERCISE - 5

Cloud Object Storage Manager Using Python & MinIO

Aim:

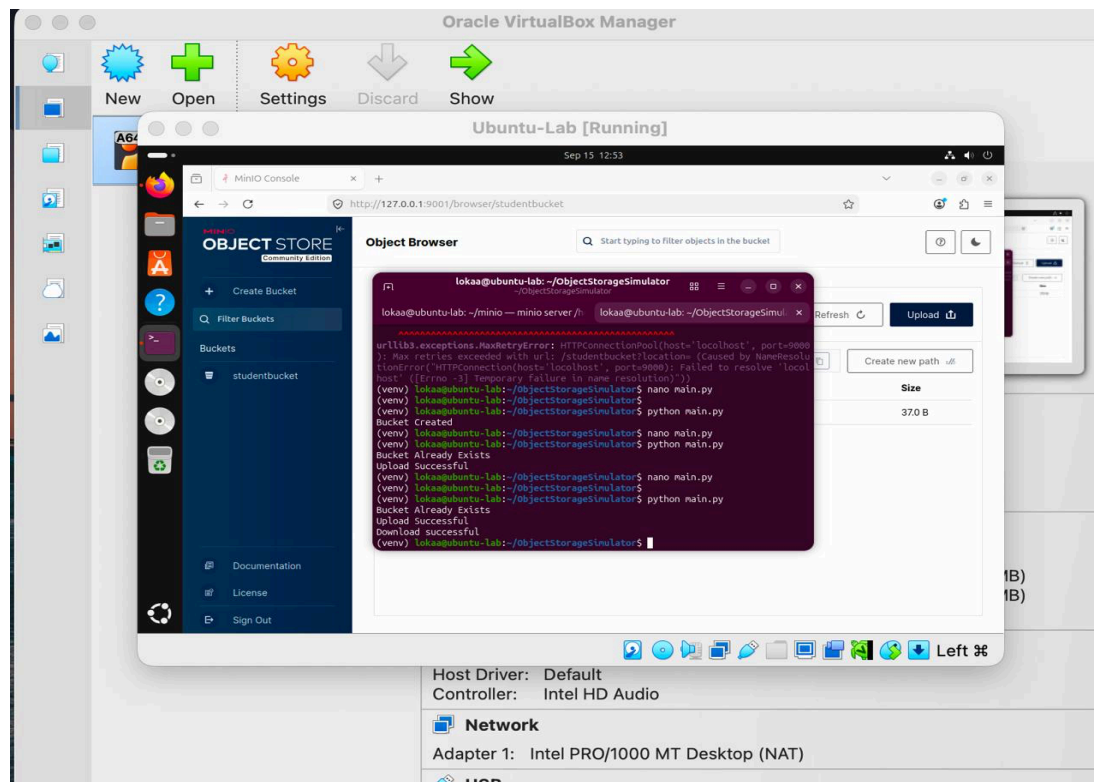
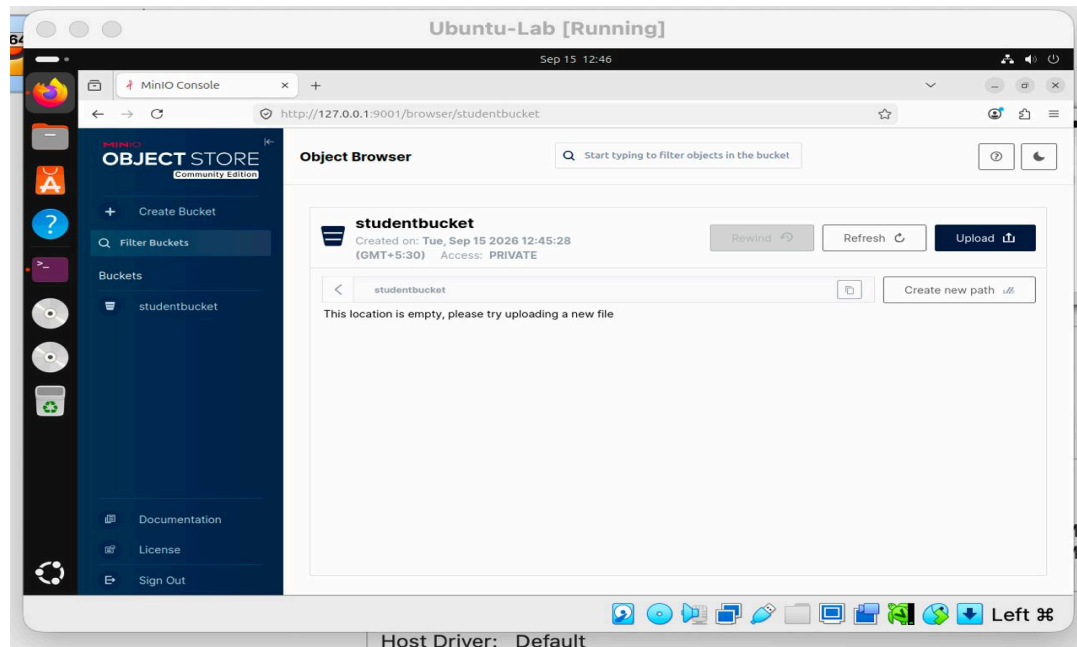
To develop a local cloud object storage manager using Python and MinIO to perform S3-compatible object storage operations such as bucket creation, object upload, download, listing, and deletion.

Algorithm:

1. Start the Ubuntu virtual machine and prepare the Python environment.
2. Install and configure the MinIO server and create a local directory for object storage.
3. Start the MinIO server and access the MinIO Object Store console through the browser.
4. Create a Python virtual environment and install the MinIO Python SDK.
5. Establish a connection between the Python application and the local MinIO server.
6. Create a bucket named `studentbucket` if it does not already exist.
7. Upload a sample file into the bucket using a unique object name.
8. Download the stored object from the bucket to the local system.
9. List the objects available in the bucket along with metadata such as object name, size, and last-modified time.
10. Delete the required object from the bucket.
11. Combine all operations into an interactive menu-driven Cloud Object Storage Manager.
12. Test each menu option and exit the application successfully.

Output:

The Cloud Object Storage Manager displayed an interactive menu with options to create a bucket, upload an object, download an object, list bucket contents, delete an object, and exit. All operations were executed successfully using the local MinIO server.



```
lokaa@ubuntu-lab: ~/ObjectStorageSimulator — pyt...  
~/ObjectStorageSimulator  
lokaa@ubuntu-lab: ~/minio — minio server /h...  
lokaa@ubuntu-lab: ~/ObjectStorageSimul... x  
(venv) lokaa@ubuntu-lab:~/ObjectStorageSimulator$ python main.py  
--- Cloud Object Storage Manager ---  
1. Create Bucket  
2. Upload Object  
3. Download Object  
4. List Bucket Contents  
5. Delete Object  
6. Exit  
Enter your choice: 1  
Bucket Already Exists  
  
--- Cloud Object Storage Manager ---  
1. Create Bucket  
2. Upload Object  
3. Download Object  
4. List Bucket Contents  
5. Delete Object  
6. Exit  
Enter your choice: █  
  
lokaa@ubuntu-lab: ~/ObjectStorageSimulator  
~/ObjectStorageSimulator  
lokaa@ubuntu-lab: ~/minio — minio server /h...  
lokaa@ubuntu-lab: ~/ObjectStorageSimul... x  
--- Cloud Object Storage Manager ---  
1. Create Bucket  
2. Upload Object  
3. Download Object  
4. List Bucket Contents  
5. Delete Object  
6. Exit  
Enter your choice: 5  
Object Deleted Successfully  
  
--- Cloud Object Storage Manager ---  
1. Create Bucket  
2. Upload Object  
3. Download Object  
4. List Bucket Contents  
5. Delete Object  
6. Exit  
Enter your choice: 6  
Exiting Object Storage Manager  
(venv) lokaa@ubuntu-lab:~/ObjectStorageSimulator$ █
```

Result:

Thus, a local S3-compatible Cloud Object Storage Manager was successfully implemented using **Python and MinIO**, and the required bucket and object storage operations were performed successfully.